We begin with a Gaussian input field amplitude in the near field:

$$A_{\text{in}}(\mathbf{r}) = \exp\left(-\frac{|\mathbf{r}|^2}{w_0^2}\right). \tag{1}$$

Its Fourier transform is also a Gaussian:

$$\mathcal{F}\{A_{\text{in}}(\mathbf{r})\}(\mathbf{k}) = \exp\left(-\frac{w_0^2|\mathbf{k}|^2}{4}\right). \tag{2}$$

—

Now, if the input field is shifted by $\mathbf{r}_0$, this corresponds to convolution with a Dirac delta:

$$A'_{\text{in}}(\mathbf{r}) = \big(A_{\text{in}} * \delta(\mathbf{r} - \mathbf{r}_0)\big)(\mathbf{r}) = A_{\text{in}}(\mathbf{r} - \mathbf{r}_0). \tag{3}$$

In the Fourier plane this results in multiplication by a linear (blazing) phase:

$$\mathcal{F}\{A'_{\text{in}}(\mathbf{r})\}(\mathbf{k}) = \mathcal{F}\{A_{\text{in}}(\mathbf{r})\}(\mathbf{k})\, e^{-i\mathbf{k}\cdot\mathbf{r}_0}. \tag{4}$$

—

Next, we apply a phase mask corresponding to a bottle beam:

$$A_{\text{mask}}(\mathbf{r}) = A'_{\text{in}}(\mathbf{r})\, e^{i\phi_{\text{bb}}(\mathbf{r})}. \tag{5}$$

The Fourier transform of this field is given by a convolution:

$$\mathcal{F}\{A_{\text{mask}}(\mathbf{r})\}(\mathbf{k}) = \Big[\mathcal{F}\{A'_{\text{in}}(\mathbf{r})\}(\mathbf{k})\Big] * \Big[\mathcal{F}\{e^{i\phi_{\text{bb}}(\mathbf{r})}\}(\mathbf{k})\Big]. \tag{6}$$

—

Thus, the final output field in the Fourier plane can be written as

$$A_{\text{out}}(\mathbf{k}) = \Big[G(\mathbf{k})\, e^{-i\mathbf{k}\cdot\mathbf{r}_0}\Big] * B(\mathbf{k}), \tag{7}$$

where

$$G(\mathbf{k}) = \exp\Big(-\tfrac{w_0^2|\mathbf{k}|^2}{4}\Big), \quad B(\mathbf{k}) = \mathcal{F}\{e^{i\phi_{\text{bb}}(\mathbf{r})}\}.$$

In words: the output is the Gaussian spot (shifted by a blazing phase) convolved with the Fourier representation of the bottle-beam mask.